

Kartik Agrawal

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Aachen, Germany

EDUCATION

RWTH Aachen University

M.Sc. Robotic Systems Engineering (Grade: 2.1)

Aachen, Germany

Oct. 2023 – Present

BITS Pilani, Pilani Campus

B.Eng. Mechanical Engineering (Grade: 7.83/10)

Pilani, India

Aug. 2017 – July 2021

Relevant coursework: Advanced Machine Learning, Computer Vision, Simulation of Robotic Systems, Sensors and Environment, Reinforcement Learning, Mechatronics, Mobile Robotics, Industrial Logistics

RESEARCH EXPERIENCE

Institute of Automatic Control (IRT), RWTH Aachen

M.Sc. Thesis

Aachen, Germany

July 2026 – Jan. 2027

- *Decentralized GNSS/Multi-sensor Factor Graph Optimization for Cooperative UAV Localization.*
- Developed a real-time GNSS simulator with live ephemeris data and configurable error models, supporting both loosely and tightly coupled architectures for comparative navigation studies.
- Implemented a discrete-time GNSS/IMU/magnetometer factor graph optimization (FGO) state estimator in GTSAM, achieving 0.3 m position RMSE validated against simulated ground truth.
- Extending the estimator with UWB inter-agent ranging and a consensus layer for cooperative localization, accounting for communication bandwidth limits, asynchronous messaging, and agent-level failure modes.
- Analyzed stock Gazebo IMU against IRT's proprietary simulation model to characterize sensor error modes and their impact on estimator performance.
- Built an ArduPilot SITL/ROS2 simulation environment with teleoperation, and a ROS2–React interface for state visualization and diagnostics.

INDEPENDENT ROBOTICS PROJECTS

Flow-Matching UAV Autonomy | *PyTorch, MuJoCo, Gazebo* | [Code](#) | [Demo](#)

Aug. 2026 – Present

- Built a Diffusion Transformer-style conditional flow-matching policy trained by imitation learning from a classical PD controller, verified on UAVs across MuJoCo and Gazebo.
- Designed the full pipeline: simplified MuJoCo dynamics, synthetic data generation, sliding-window dataset construction, AdaLN-conditioned transformer architecture, and closed-loop Euler-sampling inference.
- In progress: vision-conditioned policy for target following with a Siamese tracking head and active gimbal control.

UAV Autonomy Stack | *ROS2, PX4 SITL, Gazebo, Nav2* | [Code](#) | [Demo](#)

Feb. 2026 – Present

- Built a full-stack autonomous UAV system in ROS2/PX4 supporting manual gamepad control and perception-driven autonomous flight in offboard mode.
- Implemented ArUco-based visual localization with PD closed-loop velocity control for waypoint navigation and precision landing.
- Integrated LiDAR-based collision avoidance with real-time velocity filtering in obstacle-facing directions.
- Adapted Nav2 for UAV dynamics, enabling autonomous path planning and goal navigation on a drone platform.
- Built a C++ marker-map annotator for semantic map annotation, feeding VLM-based reasoning into the estimation and control layers.

Deep Learning Perception | *PyTorch, MoveIt*

Feb. 2026

- Trained YOLOv8 detectors and ResNet18 classifiers for industrial inspection tasks at the Europe Embodied and Intrinsic AI hackathons.
- Implemented MoveIt pick-and-place in simulation using Franka Panda and Elephant Robotics MyCobot 280.

PROFESSIONAL EXPERIENCE

T-Systems International

Software Engineering Intern

Aachen, Germany

Sept. 2024 – Jan. 2026

- Designed and implemented a Timefold AI optimization engine for Rego policy-driven automated scheduling, with tuned hard and soft constraints.
- Developed supporting tooling that reduced solver runtime and constraint violations, plus a web interface for results visualization.
- Built a containerized microservices architecture for real-time data streaming using Redpanda, custom Go plugins, PostgreSQL persistence, and AMQP for reliable event processing.
- Instrumented the system for observability using Grafana and Tempo.

LTIMindtree

Full-Stack Software Engineer

Bangalore, India

July 2021 – Aug. 2023

- Developed production microservices and full-stack applications for client-facing projects using Spring Boot and the MERN stack.
- Took multiple proofs of concept from problem definition through demonstration to client approval and deployment.
- Worked with relational and graph databases on automation-focused solutions.

Amazon India (ASSPL)

Area Management Intern, Fulfillment Center

Mumbai, India

Feb. 2021 – June 2021

- Identified gaps and sources of defects in the inbound phase of shipments through operational area surveying and analysis of process and defect metrics.
- Developed and deployed VBA macros for package sorting and tracking, automating shipment-wise pallet sorting and identification through shipment coded pallet labels.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript (MERN), SQL

Navigation & Estimation: GNSS (GPS, Galileo), inertial navigation, IMU, magnetometer, UWB ranging, multi-sensor fusion, factor graph optimization, GTSAM

Simulation & Robotics: ROS2, Gazebo, MuJoCo, PX4, ArduPilot, Nav2, MoveIt, Isaac Sim, CAD

Machine Learning: Imitation learning, transformers, VLMs, agentic AI, PyTorch

Tools: Git, Docker, Linux, Redpanda, PostgreSQL, MySQL, Grafana

LANGUAGES

English (Fluent) | German (A2) | Hindi (Native)